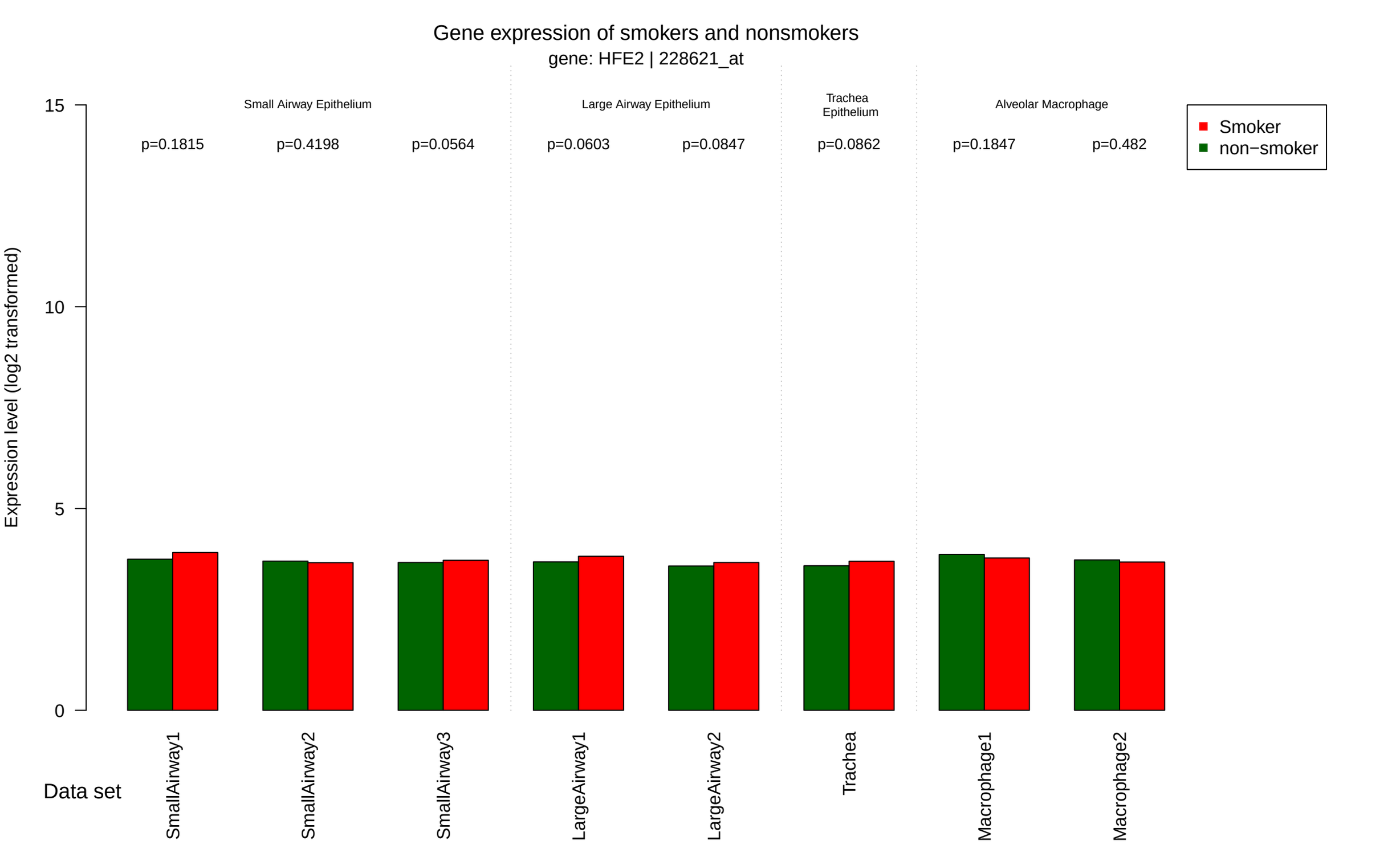
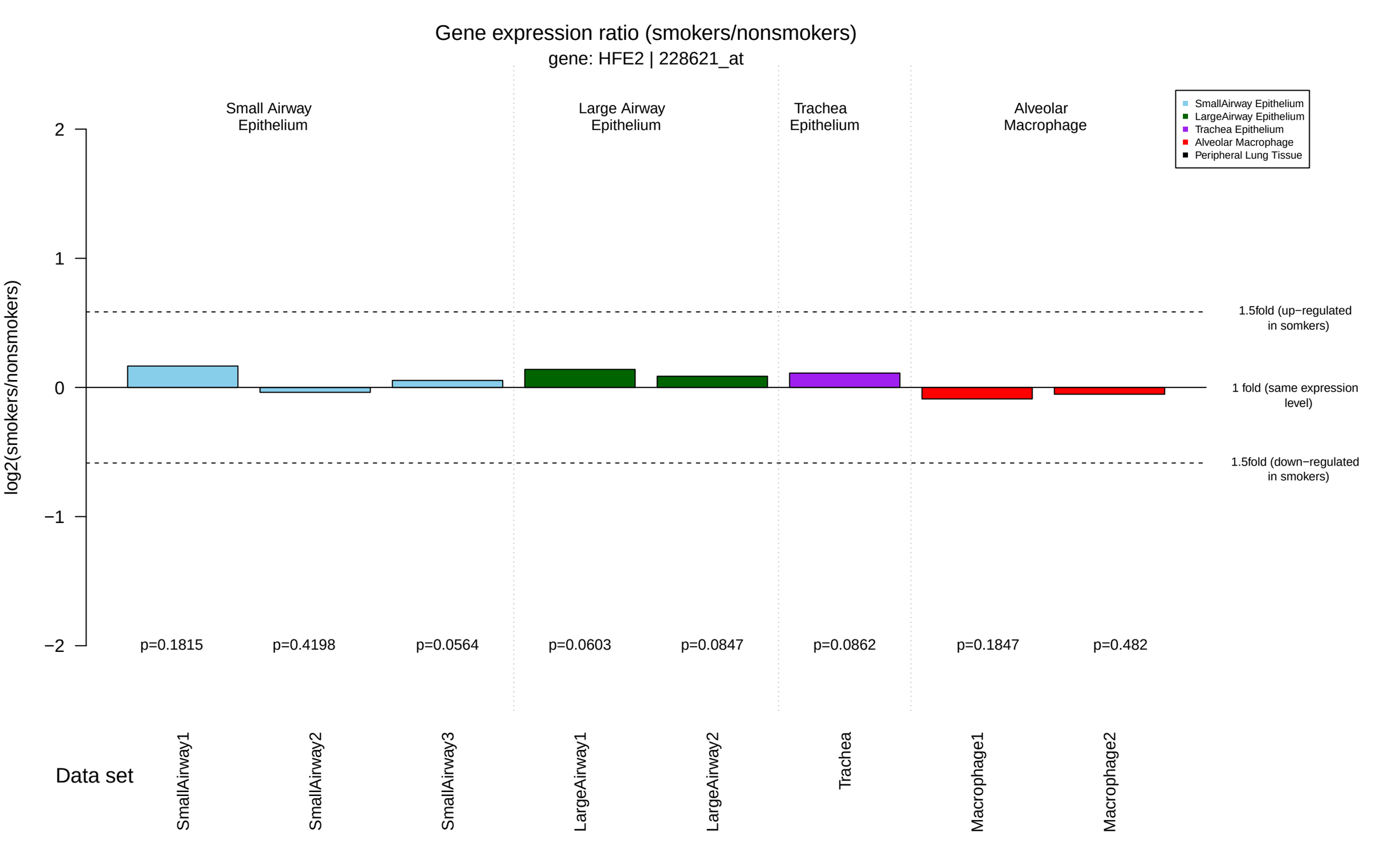
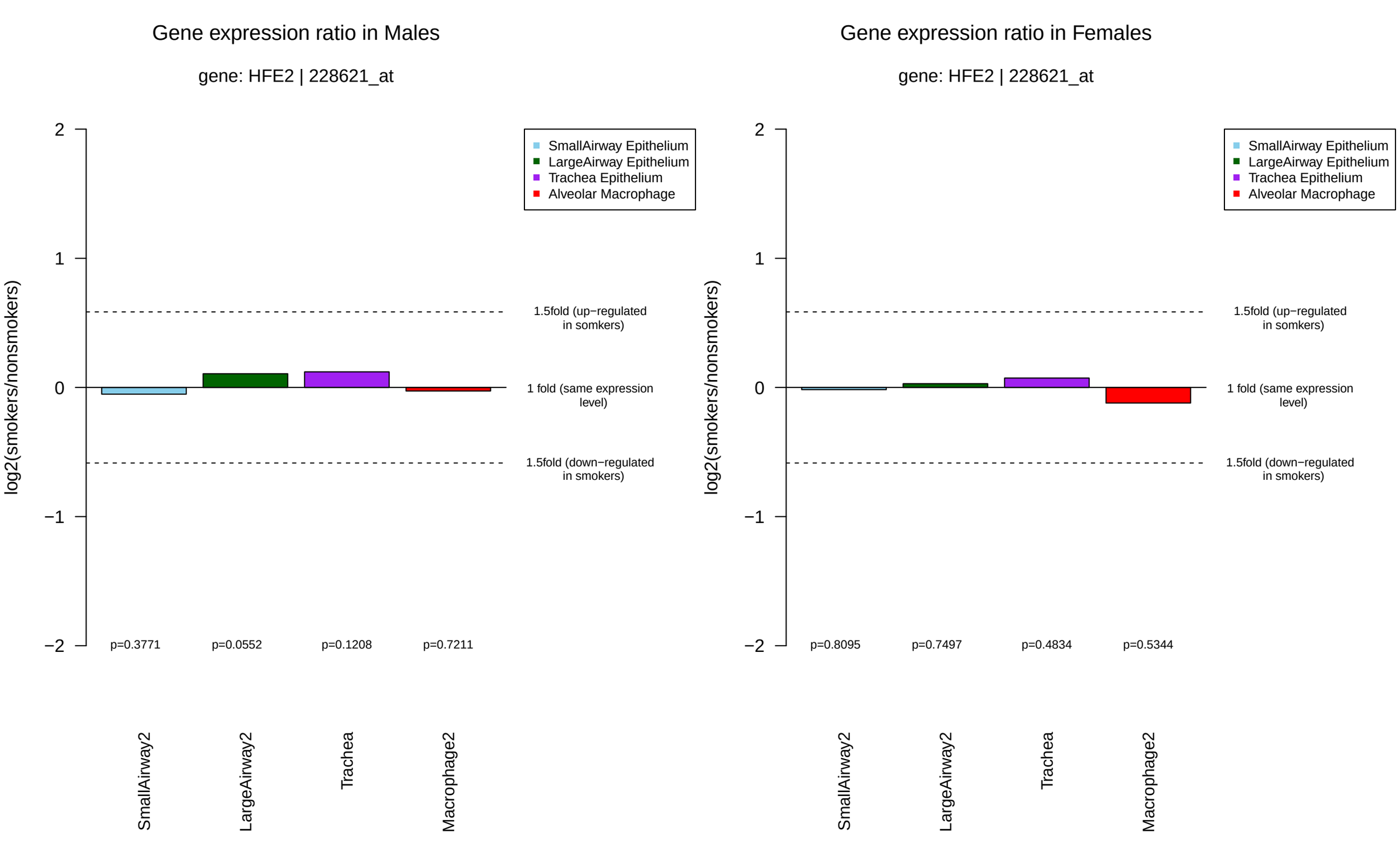
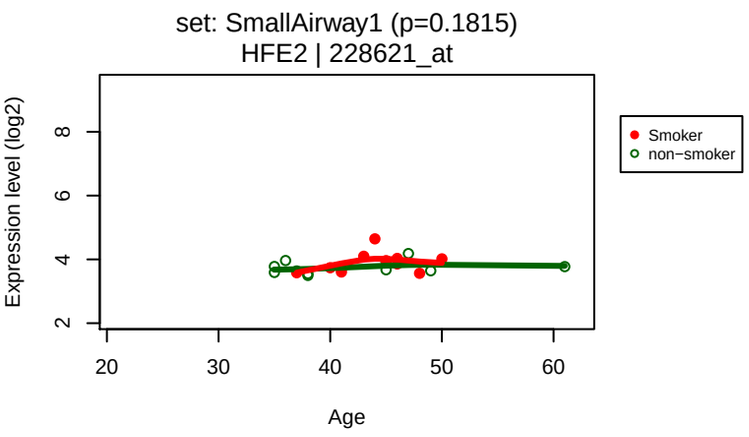
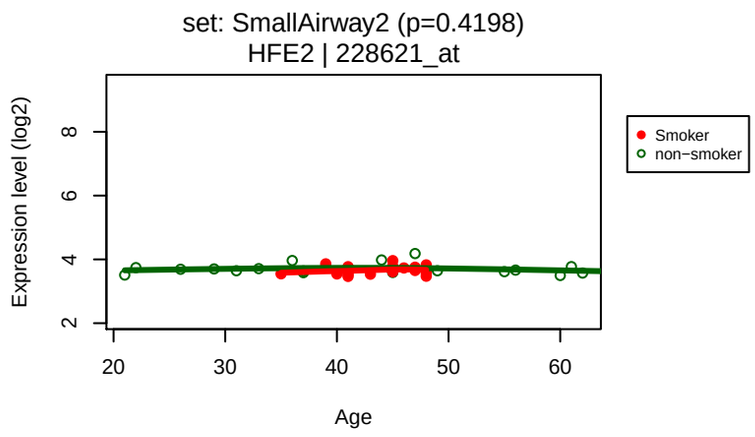
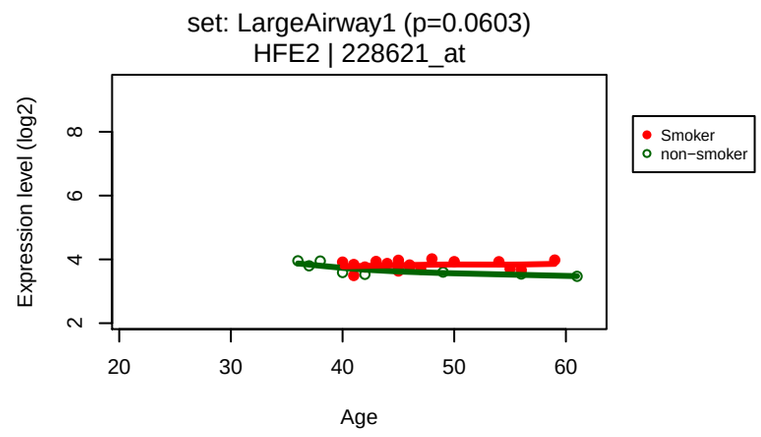

Gene expression of smokers and nonsmokers
gene: HFE2 | 228621_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1815
p=0.4198
p=0.0564
p=0.0603
p=0.0847
p=0.0862
p=0.1847
p=0.482
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

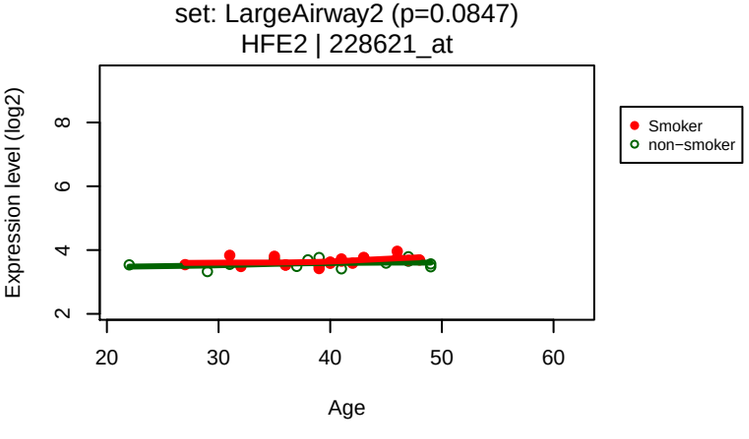

Gene expression ratio (smokers/nonsmokers)
gene: HFE2 | 228621_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1815
p=0.4198
p=0.0564
p=0.0603
p=0.0847
p=0.0862
p=0.1847
p=0.482
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

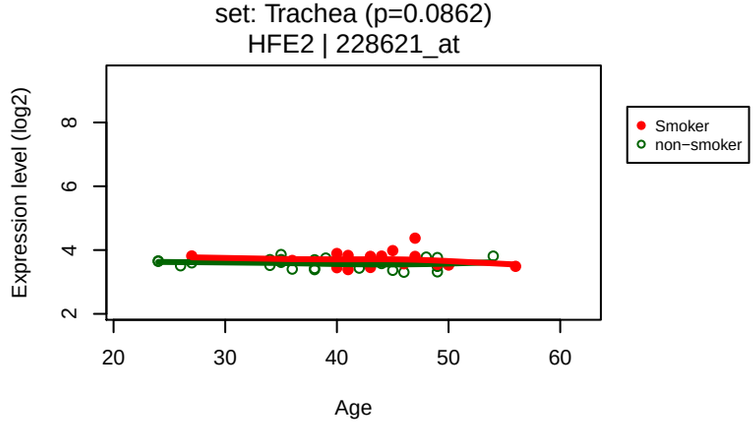

Gene expression ratio in Males
gene: HFE2 | 228621_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3771
p=0.0552
p=0.1208
p=0.7211
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HFE2 | 228621_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8095
p=0.7497
p=0.4834
p=0.5344
SmallAirway2
LargeAirway2
Trachea
Macrophage2

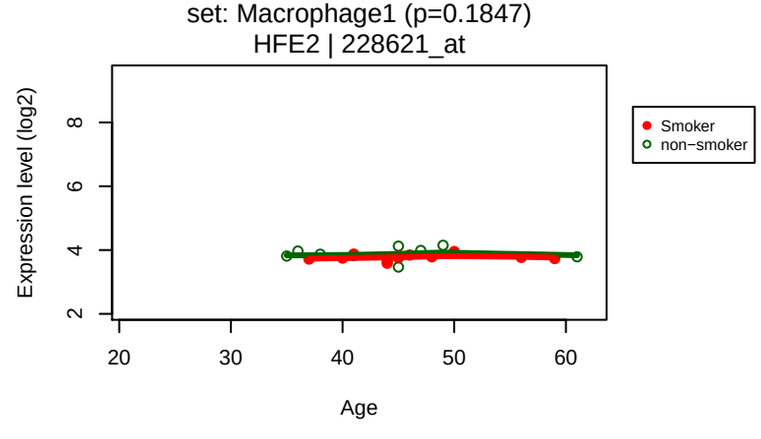

set: SmallAirway1 (p=0.1815)
HFE2 | 228621_at
Expression level (log2)
Age
Smoker
non−smoker

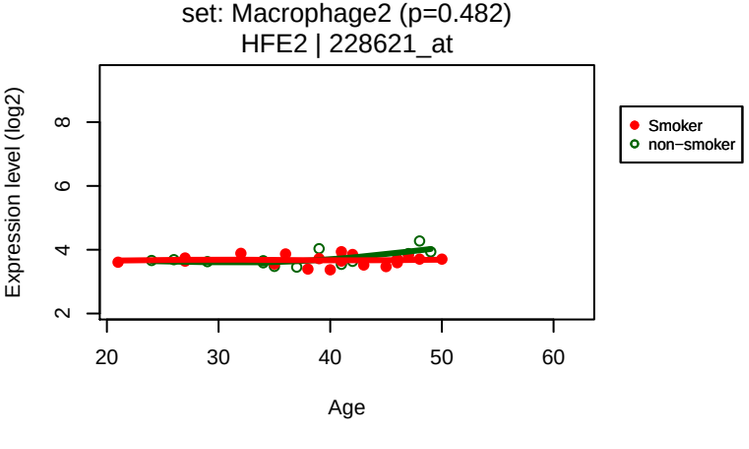

set: SmallAirway2 (p=0.4198)
HFE2 | 228621_at
Expression level (log2)
Age
Smoker
non−smoker

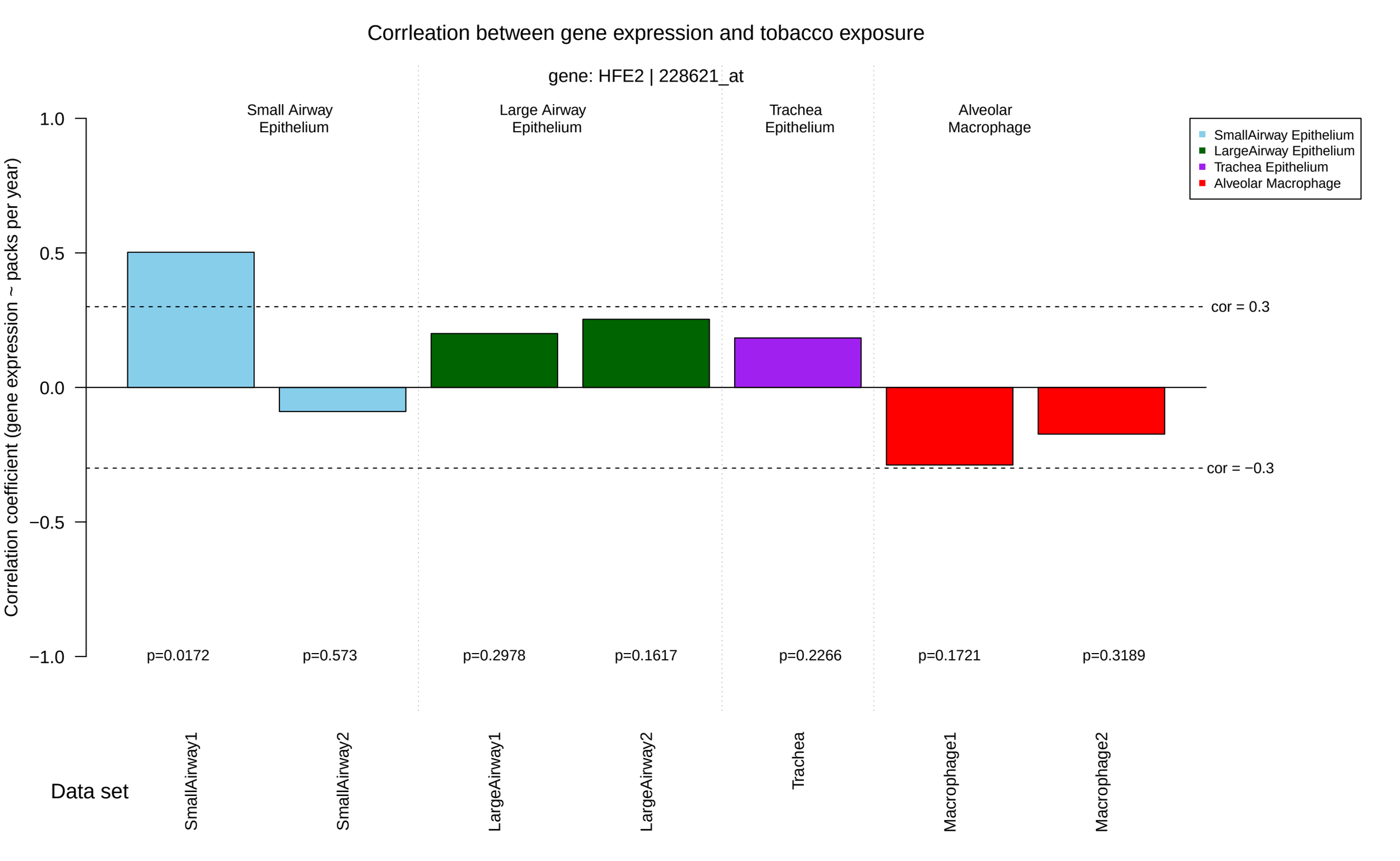

set: LargeAirway1 (p=0.0603)
HFE2 | 228621_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0847)
HFE2 | 228621_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0862)
HFE2 | 228621_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1847)
HFE2 | 228621_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.482)
HFE2 | 228621_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HFE2 | 228621_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0172
p=0.573
p=0.2978
p=0.1617
p=0.2266
p=0.1721
p=0.3189
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2